Resource coupling and verification in team continuation

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Abstract

We study how resource interdependence shapes the value of verification before a team decides whether to continue. Two participants privately know their continuation values and their exposure to project resources. An organizer can verify the resource quantities and settle the associated commitments. Settlement reduces resource-based incentives to imitate a committed partner and releases capacity for further investment. Different settlement levels across the two participants require costly repair of a shared interface. We characterize exact reciprocal continuation and solve the joint choice of settlement and adaptive verification. At the separable benchmark, the two verification values are additive. Their local interaction as coupling increases equals the product of the average settlement changes induced by individual verification. Resource coupling can therefore generate either complementarity or substitution, depending on the direction of those adjustments. Verification can also induce the organizer to modularize the project. Allowing continuation errors changes the preferred timing of information: within a specified class of cutoff equilibria, early disclosure can outperform the delay favored by exact selection. A productive-investment extension shows how joint output raises imitation returns. The results connect project architecture to the organization of information acquisition and the cost of voluntary cooperation.

Keywords: Costly verification; team continuation; resource coupling; information timing; mechanism design; modularity.

JEL classification: D82, D83, D86, L23.

1 Introduction

Consider two collaborators deciding whether to continue after a trial project. One wants to preserve the partnership; the other may chiefly value the resources that remain available through it. Both can express willingness to continue by making a costly investment. An organizer can make that investment easier to interpret by checking the resource rights behind it. Yet using the information may require changing the project itself. If the two resource modules share an interface, releasing one person's commitments while retaining the other's creates an adjustment cost. When should the organizer check the participants separately, and when is it worth checking both?

The question links the value of information to an organization's capacity to adapt. A resource check can justify less settlement because it removes uncertainty about the strongest potential imitator. It can also justify more settlement when independently usable resources are especially valuable. With separable modules, each response can be chosen on its own. With a shared interface,

one person's adjustment changes the cost of the other person's adjustment. The paper studies this connection between resource coupling and the joint value of verification.

The signaling literature provides the starting point. Costly actions can reveal intentions to invest in a relationship, including when both participants are privately informed (Camerer 1988). Resource benefits complicate that inference: external incentives change the interpretation of behavior (Bénabou and Tirole 2006), and multiple private attributes can affect the same observed action (Frankel and Kartik 2019). We accordingly distinguish a participant's value for the current team from the objective resource quantity attached to continued cooperation. A resource check reveals the latter. Further investment must still convey the former.

Costly-verification theory asks how checking information affects an incentive-compatible allocation (Ben-Porath et al. 2014; Erlanson and Kleiner 2020). Certification models also connect information acquisition to the screening of privately held information (Asseyer and Weksler 2024; Celik and Strausz 2026). Here, verification precedes a resource-settlement decision that changes subsequent signaling incentives. The organizer commits to that decision and to the verification procedure; each participant retains final consent over team continuation. Thus the information is valuable through the resource arrangement it permits and the voluntary behavior that arrangement supports.

Project architecture determines how these adjustments interact. Each module can be settled separately, but unequal settlement levels require a repair of their common interface. A quadratic repair cost connects separable resources to the limit of common settlement. The specification builds on the distinction between modular adaptation and coordination across interdependent activities (Baldwin and Clark 2000; Dessein and Santos 2006). The main result identifies the resulting interaction between two resource checks. At zero coupling, their values add. The derivative of their interaction with respect to coupling equals the product of their average settlement changes. Checks that move settlement in the same direction are locally complementary; opposite movements make them local substitutes. This gives an adjustment-based expression for the information interaction studied by Börgers et al. (2013).

The formula explains why verification can change the preferred project architecture. Suppose each check would allow less settlement on average. Checking just one participant tends to leave the modules at different settlement levels. Checking the second lets both modules adjust, which increases the joint value relative to the two stand-alone checks. The same value of adapting to resource records can make it worthwhile to pay for a more modular project. With linear redesign costs, the organizer chooses between full modularization and no redesign. Asset-control theories study how rights over productive assets affect incentives (Grossman and Hart 1986; Hart and Moore 1990); the architectural choice here concerns the cost of adjusting resource commitments while cooperation remains voluntary.

The best verification procedure need not check both participants together. After a first report, the organizer may learn that another check would permit enough additional adaptation to justify its cost. We solve all eight distinct pure procedures and their mixtures. A conditional procedure can be optimal at moderate coupling, while paired checking emerges near the common-settlement limit. Convex costs of realized workload can favor conditional checking even while paired checks retain an information advantage. This separation of the informational benefit from the cost of providing it is central to the interpretation of verification institutions (Dranove and Jin 2010).

The continuation target determines how much information is worth obtaining. Under exact reciprocal continuation, every pair with high continuation values stays together and every other

pair stops. A resource report with positive error retains extreme resource states among the possible imitators. At the baseline, it therefore leaves the exact-selection constraint unchanged. Allowing some errors permits less costly allocations. We optimize a class of cutoff equilibria that preserves the final refusal decision and evaluate erroneous continuation separately from erroneous dissolution. The calculation provides feasible policies and matching bounds for that class.

Information timing changes the same investment incentives. A favorable early record about one collaborator makes that person more attractive to a low-value counterpart, which raises the return to imitation. With limited investment capacity, exact selection can then require more resource settlement. Permitting errors allows the organizer to use early records for different continuation decisions, and the preferred timing can reverse. The comparison connects information design (Kamenica and Gentzkow 2011; Bergemann and Morris 2019) to the incentive effects of public information in relationships (Kolotilin and Li 2021). Removing the capacity constraint identifies the source of the delay advantage: early disclosure can then reproduce the value of every delayed policy.

Trial collaboration supplies an organizational interpretation of the environment. Research on entrepreneurial team formation distinguishes assembling a group from organizing its subsequent cooperation (Lazar et al. 2020); founder equity agreements are one way to allocate rights within that group (Hellmann and Wasserman 2017). Starting-small models explain how initial interaction and commitment support later cooperation (Rauch and Watson 2003; Watson 1999). We focus on a fixed pair at the continuation stage, when the resources left by the trial affect investment incentives. A productive-investment extension lets those investments also create joint output. Because a mimicking low type receives that output too, productivity changes the investment needed for selection.

Related working papers examine other ways to shape a signaling environment. *Signaling Design* by Camboni et al. (2025, arXiv:2502.02328) studies monitoring and fees, while Han and Sam (2024, arXiv:2402.14003) analyze multidimensional signaling under a resource constraint. Moriya and Stern's *Legal Commitment as Costly Signaling* (2026, UTMD Working Paper 137) studies commitment with private relationship values. Our focus is the adjustment link between two resource checks. The analysis proceeds from continuation incentives to optimal settlement, then asks how verification and modularization use that link. Error tolerance changes the allocation being implemented; productive investment changes the cost of implementing it.

2 Environment and voluntary continuation

2.1 Participants and private values

There are two participants, indexed by $i \in \{A, B\}$, with counterpart j. Each knows a private continuation type and an objective resource exposure:

$$\theta_i \in \{H, L\}, \qquad \Pr(\theta_i = H) = p_i \in (0, 1), \qquad v_H > v_L > 0, \tag{1}$$

$$r_i \in \{r_-, r_+\}, \qquad 0 < r_- < r_+, \qquad \Pr(r_i = r_+) = q_i \in (0, 1). \tag{2}$$

Continuation types are independent across participants. Resource exposures are also independent across participants and independent of continuation types. The exposure r_i is a checkable quantity or entitlement. This independence isolates the channel of interest: verification informs the organizer

about the incentive to retain project resources, while the participant's value for continued cooperation remains private.

Each participant values a counterpart's willingness to continue. Write

$$g(H) = \eta > 0, \quad g(L) = -\ell < 0, \qquad a_H = v_H + \eta, \quad a_L = v_L + \eta, \quad \Delta = v_H - v_L. \tag{3}$$

The regularity condition is

$$\ell > v_H + br_+, \tag{R}$$

where $b > 0$ is the value of one unit of resource exposure. Both private types prefer to continue with a known high type, and both prefer to stop with a known low type. A low type therefore wants to imitate a high type when imitation attracts a high-value counterpart. Selection is needed even though the low type would accept such a partnership voluntarily.

2.2 Settlement and interface repair

The organizer can settle each resource module separately. Write $\mathbf{x} = (x_A, x_B)$ for the settlement vector. Each choice is $x_i \in [0, 1]$, and the module's own settlement cost is $\kappa x_i^2/4$, with $\kappa > 0$. If the two modules are settled to different degrees, the shared interface requires repair. A repair quantity d must satisfy $d \geq x_A - x_B$, $d \geq x_B - x_A$ and $d \geq 0$. Repair costs $\lambda d^2/2$, with $\lambda \geq 0$. Cost minimization gives $d = |x_A - x_B|$, so total settlement cost is

$$C_\lambda(x_A, x_B) = \frac{\kappa}{4}(x_A^2 + x_B^2) + \frac{\lambda}{2}(x_A - x_B)^2. \tag{4}$$

Each participant pays the cost of their own module and half the interface cost. For example, settling A's module while leaving B's module unchanged requires reconciling the different resource arrangements at their common interface. The parameter λ measures the cost of that reconciliation. At zero coupling, the modules can be adapted separately; high coupling makes a common settlement increasingly attractive. Equation (4) gives this coordination-versus-adaptation trade-off an explicit technological form (Milgrom and Roberts 1990).

Settlement also changes how resources can be used. The fraction ζx_i becomes available independently of team continuation. If the team continues, the fraction χx_i can subsequently be restored to its relationship-bound use. Assume

$$\zeta \geq 0, \quad \chi \geq 0, \quad \zeta + \chi \leq 1, \qquad \gamma = 1 - \chi, \quad t_i(x_i) = 1 - \gamma x_i. \tag{5}$$

At the restoration node, A proposes restoration and B decides whether to consent. Each receives $\chi x_i r_i b \geq 0$, with no new cost or private information. Restoration is sequentially optimal; ties are resolved in favor of restoration. Thus t_i already incorporates the actual resource recovery. For continuation decision $D \in \{0, 1\}$, the resource account satisfies

$$\zeta x_i + D t_i(x_i) \leq 1. \tag{6}$$

The account distinguishes resources released for independent use from those available through continued cooperation. Future restoration is already included in the residual coefficient t_i. Specifying

that recovery matters for commitment, since later recontracting can change the payoff consequences of a current arrangement (Vasconcelos 2017; Halac and Yared 2020).

2.3 Investment and information

Participants invest simultaneously after settlement. Investment is measured in units of real opportunity cost and satisfies

$$0 \le y_i \le K_i(x_i) = k_{0i} + k_{1i}x_i, \qquad k_{0i} > 0, \quad k_{1i} \ge 0. \tag{7}$$

Unsettled commitments occupy part of the activity window in which a participant can demonstrate willingness to continue. Settlement therefore works on both sides of the signaling constraint: it lowers resource-based imitation returns and releases room for investment. The cost of investment is sunk before final consent. The main analysis uses purely costly investment, in the tradition of Spence (1973); Section 6 adds a joint production return.

A binary record about each continuation type has precision ε:

$$\Pr(S_i = + \mid H_i) = \varepsilon, \qquad \Pr(S_i = + \mid L_i) = 1 - \varepsilon, \qquad \tfrac{1}{2} \le \varepsilon < 1. \tag{8}$$

Conditional on types, the records are independent. The organizer initially holds them; participants have no private advance observation of their own record. Each record can be released before investment or after investment but before final consent. A release schedule $R \in \{00, 10, 01, 11\}$ records whether A's and B's records are early. A delayed record cannot be used secretly to select an observable audit or settlement policy.

For an early record, the posterior is

$$p_i^+ = \frac{p_i\varepsilon}{p_i\varepsilon + (1-p_i)(1-\varepsilon)}, \qquad p_i^- = \frac{p_i(1-\varepsilon)}{p_i(1-\varepsilon) + (1-p_i)\varepsilon}. \tag{9}$$

Without an early record, the corresponding probability remains p_i. Independence across participants is preserved conditional on early public records.

Resource checks report the objective exposure r_i. A report is correct with probability $\rho \in [1/2, 1]$, independently across participants conditional on true exposures. Reports are automatically public. At $\rho = 1$, a report identifies the resource quantity. For $\rho < 1$, it changes the probabilities of the two quantities while leaving both possible. We use this measurement kernel to study how verification precision affects screening, a related concern in Ball and Kattwinkel (2025).

2.4 Adaptive verification and its cost

An organizer can check each participant at most once. It may stop after the first check or make the second check depend on the first report. The eight distinct pure programs are

$$\mathcal{T} = \{00, 10, 01, 11, A_-, A_+, B_-, B_+\}. \tag{10}$$

Program 00 checks nobody; 10 and 01 check only the named participant; 11 checks both. Program A_- checks A first and checks B only after a low report. The other conditional programs follow the

corresponding rule. The order of two unconditional checks has the same final information and cost, so it is represented once. Public mixtures of these programs include randomized stopping after the first report.

Let N be the realized number of checks and $a = E[N/2 \mid h]$ the coverage purchased after early history h. The real service cost and workload cost are

$$C_a(a) = c_a a + \frac{d_a a^2}{2}, \qquad C_N(N) = \frac{\nu}{2}(N/2)^2, \quad c_a \geq 0,\ d_a > 0,\ \nu \geq 0. \tag{11}$$

The service fee purchases coverage before the public program lottery; the workload fee charges for the number of checks actually performed. Thus two procedures can buy the same average coverage and incur different realized workload costs. This cost distinction enters the design problem alongside the informational benefit of each check (Townsend 1979; Border and Sobel 1987).

2.5 Payoffs and equilibrium

After solving the restoration node, the common payoff specification is

$$\begin{aligned} u_i = Q_i - \frac{\kappa}{4}x_i^2 - \frac{\lambda}{4}(x_A - x_B)^2 - \frac{C_a(a)}{2} - \frac{\nu}{4}(N/2)^2 \\ + \zeta x_i r_i b + D\big[v_{\theta_i} + g(\theta_j) + t_i(x_i) r_i b\big] - y_i + \frac{\varphi}{2} y_A y_B. \end{aligned} \tag{12}$$

In equation (12), u_i denotes participant utility. The production parameter $\varphi \geq 0$ permits investment to produce $\varphi y_A y_B$, divided equally. This output is realized before final consent and remains available when the team stops. The main analysis takes $\varphi = 0$. All resource gains and costs are measured in the same utility units.

The organizer commits to a public policy before private types are drawn. A then decides whether to enter a trial project; B observes the invitation and decides whether to join. After the indivisible trial, each earns net output Q_i. The organizer releases the early records. It then conducts the prescribed checks and settles the resource modules. Participants invest simultaneously. They then observe both investments and the remaining records. A offers binding consent to continuation, and B accepts or refuses. Restoration follows continuation.

Direct cooperation remains available after a rejected trial. We model it as a bounded outside-opportunity class with the same kind of investment and one record of the same precision. It has no trial output or project-interface repair obligation. Its total resource value is bounded by br_+, and its capacity can be as high as $\max_i(k_{0i}+k_{1i})$. Continuation utility is therefore bounded by $M_i = a_H + br_+$. The entry result below uses this bound uniformly across admissible direct-cooperation equilibria.

Participants use perfect Bayesian equilibrium (PBE). Actions are optimal at each information set. Positive-probability histories obey Bayes' rule; beliefs are also specified at unused actions. This is the assessment-based equilibrium framework of Fudenberg and Tirole (1991); Kreps and Wilson (1982) provide the additional consistency requirement for sequential equilibrium.

The initial design target is exact reciprocal continuation:

$$D = \mathbf{1}_{\{\theta_A = H, \theta_B = H\}}. \tag{HH}$$

It must hold for every resource state with positive probability at each public history. Write $\mathcal{E}_{HH}(\mathcal{D})$ for the corresponding PBE set under policy $\mathcal{D}$. With $W = E[u_A + u_B - Q_A - Q_B]$, the exact design value is

$$V_0 = \sup_{\mathcal{D}} \sup_{e \in \mathcal{E}_{HH}(\mathcal{D})} W(e). \tag{13}$$

The design objective is weak implementation: the organizer chooses the least costly equilibrium that delivers reciprocal continuation. This establishes the cost of achieving a given allocation. Section 5 then lets the organizer trade that cost against continuation errors.

3 Implementation under resource coupling

3.1 The cost of reciprocal continuation

To interpret an investment, the counterpart must know which types would find it worthwhile. Fix a public history z immediately before investment. Write $p_i(z)$ for the high-type probability, and describe resource exposure by its support endpoints $\underline{r}_i(z), \overline{r}_i(z)$ and mean $m_i(z)$. An error-free check makes this support a singleton; a noisy report changes the mean while retaining both endpoints.

The strongest imitator determines the minimum signal cost. Define

$$\begin{aligned} L_i(z, x_i) &= p_j(z)\big[a_L + t_i(x_i) b \overline{r}_i(z)\big], \\ H_i(z, x_i) &= p_j(z)\big[a_H + t_i(x_i) b \underline{r}_i(z)\big]. \end{aligned} \tag{14}$$

The lower bound comes from the low type most willing to imitate, who has the largest resource exposure. The upper bound comes from the high type least able to justify the cost, who has the smallest exposure. Both expect a valuable relationship only when the counterpart is high. The probability p_j therefore scales both bounds, while the resource spread can put them in the wrong order.

Proposition 1 (Exact implementation)**.** *Suppose condition (R) holds, the public history has full continuation-type support, and investment is purely costly. Exact reciprocal continuation is implementable in PBE if and only if, for each participant,*

$$L_i(z, x_i) \leq \min\{K_i(x_i), H_i(z, x_i)\}. \tag{15}$$

In every equilibrium implementing this allocation, low types invest zero. For each participant, all resource subtypes with high continuation value use a common positive cost in the interval

$$k_i \in [L_i(z, x_i), \min\{K_i(x_i), H_i(z, x_i)\}]. \tag{16}$$

The interval separates two reasons why reciprocal continuation may be costly to implement. The condition $L_i \leq H_i$ requires genuine willingness to continue to outweigh the largest resource-based motive to imitate. The condition $L_i \leq K_i$ requires enough investment capacity to deter that imitator. Verification can remove the first conflict by identifying resources, while settlement can relax both constraints. Resource subtypes with high continuation value use a common signal because they

can copy one another's complete plans and obtain the same continuation allocation at a lower cost. Appendix A.1 gives the proof.

The low type's final consent is part of the imitation calculation. After copying the positive signal, that type wants to continue with a recognized high counterpart. Its return must be offset by the signal cost. A late record with full support leaves the path's pure inference unchanged, so it cannot eliminate this deviation.

3.2 Optimal settlement at a public history

The investment interval gives a separate lower bound for each module. Put

$$\begin{aligned} e_i &= \max\{0, e_i^s, e_i^k\}, \\ e_i^s &= \left[\frac{1-\Delta/[b(\overline{r}_i - \underline{r}_i)]}{\gamma}\right]_+, \\ e_i^k &= \left[\frac{p_j(a_L + b\overline{r}_i) - k_{0i}}{k_{1i} + p_j\gamma b\overline{r}_i}\right]_+. \end{aligned} \tag{17}$$

The sorting term e_i^s is zero for a singleton resource support. If a denominator is zero, its original incentive inequality is checked directly. In particular, $\gamma = 0$ leaves the sorting inequality unchanged by settlement. Any impossible inequality or bound above one makes the exact allocation infeasible.

Conditional welfare is obtained by adding the actual payoffs. Let

$$P_z = p_A(z)p_B(z), \qquad n_i = \overline{r}_i - m_i, \qquad \beta_i = b[\zeta m_i + P_z\gamma n_i]. \tag{18}$$

The gap n_i measures the signal burden imposed on the average high type by the most resource-exposed low type. A resource check removes this gap when it identifies exposure exactly. The coefficient β_i combines the independent-use value of settlement with its reduction of that extra signal burden.

Proposition 2 (Settlement and conditional welfare)**.** *When the individual inequalities are feasible, the exact implementation set is the rectangle* $[e_A, 1] \times [e_B, 1]$*. Its highest conditional welfare, before verification fees and trial output, is obtained with* $k_i = L_i$ *and equals*

$$F_z(\mathbf{x}) = 2P_z\Delta - P_z b(n_A + n_B) + \beta_A x_A + \beta_B x_B - C_\lambda(\mathbf{x}). \tag{19}$$

This function has a unique maximizer on a nonempty implementation rectangle. Its negative Hessian is

$$\mathbf{Q}_\lambda = \begin{pmatrix} \kappa/2 + \lambda & -\lambda \\ -\lambda & \kappa/2 + \lambda \end{pmatrix} \succ 0. \tag{20}$$

Equation (19) separates the benefits of settlement from its adjustment costs. Releasing a module creates independent resource value and reduces the signal needed to deter imitation. The organizer balances these gains against the module's settlement cost and the mismatch created at the shared interface. The negative Hessian is positive definite, so the first-order conditions on the implementation rectangle determine a global optimum (Boyd and Vandenberghe 2004).

3.3 The local value of coupling

A check can move optimal settlement in either direction. It may remove a worst-case imitation constraint and permit less settlement. It may instead reveal resource quantities whose independent use makes more settlement desirable. The direction of each response determines whether a small interface cost connects the two checks as complements or substitutes.

Let $G_{ab}(\lambda)$, for $a, b \in \{0, 1\}$, be expected optimized welfare when the indicated participants are checked. These values are defined whenever their positive-probability records are implementable. Define

$$J(\lambda) = G_{11}(\lambda) - G_{10}(\lambda) - G_{01}(\lambda) + G_{00}(\lambda). \tag{21}$$

Theorem 1 (Local verification interaction)**.** *Fix an early public history and suppose all four verification values are defined. At the separable benchmark, $J(0) = 0$. Let X_i^0 be participant i's optimal separate settlement without a resource check. Let $X_i^1(r_i)$ be its optimal separate settlement after a check. Then*

$$J'(0+) = \big(E[X_A^1] - X_A^0\big)\big(E[X_B^1] - X_B^0\big). \tag{22}$$

If both average changes are nonzero and have the same sign, $J(\lambda) > 0$ for all sufficiently small positive λ. Opposite signs imply local substitution.

Consider checks that would make each module settle less on average. Checking only one person creates a tendency for that module to move away from the unchecked module. Checking the other permits an adjustment in the same direction, which makes the checks locally complementary. Opposite adjustment directions instead increase their relative mismatch and generate substitution. Algebraically, the envelope effect of coupling is minus one half of the squared settlement difference. The cross-difference cancels the individual squared terms and leaves the product in equation (22). Appendix A.4 applies the value-function argument of Milgrom and Segal (2002) to the information comparison in Börgers et al. (2013).

The signs in Table 1 illustrate both possibilities. In the baseline, each check lowers expected settlement, so their product is positive. In the heterogeneous environment, verification increases A's settlement and decreases B's settlement, so the interaction is negative. The direction comes from how the two modules adapt to information; coupling alone supplies no universal sign.

Table 1: Verification-induced settlement changes at the separable benchmark.

Environment	A: mean change	B: mean change	$J'(0+)$
Baseline	-0.26906907	-0.26906907	0.07239816
Heterogeneous technology	0.06313660	-0.10576000	-0.00667733

Note: Changes are expected settlement after verification minus settlement without verification. The derivative is their product.

3.4 Common settlement as a limit

When the interface is expensive to repair, information has less room to generate separate adjustments. The next result connects this finite adjustment cost to the limiting project in which both participants

must settle their resources together.

Theorem 2 (Strong-coupling limit)**.** *Fix an implementable public history. Let F^*_λ be optimal conditional welfare and F^*_∞ the optimum subject also to $x_A = x_B$. The function F^*_λ is nonincreasing and convex in λ, and converges to F^*_∞. With*

$$B_z = \max_i \max\{|\beta_i|, |\beta_i - \kappa/2|\},$$

for $\lambda > 0$,

$$0 \le F^*_\lambda - F^*_\infty \le \frac{B_z^2}{2\lambda}. \tag{23}$$

Finite verification-program values and their optimal mixtures have the corresponding limit. The interaction $J(\lambda)$ converges, without a general monotonicity restriction.

The bound follows by raising the smaller settlement to the larger one. This maintains both individual implementation constraints and eliminates the interface mismatch. The separate-settlement objective changes at most linearly in the mismatch, while the interface penalty is quadratic. In the common limit, the scalar solution is

$$x^* = \operatorname{clip}\left(\frac{\beta_A + \beta_B}{\kappa}, \max\{e_A, e_B\}, 1\right). \tag{24}$$

4 Verification and project architecture

4.1 The complete verification-program problem

The organizer chooses which information to obtain before selecting settlement. Let $G_\tau(h;\lambda)$ be the expected optimum in equation (19) across all final records of program τ. A program is infeasible if any of its positive-probability records cannot implement the target. Let $a_\tau = E_\tau[N/2]$ and $b_\tau = E_\tau[(N/2)^2]$, and set $\widetilde{G}_\tau = G_\tau - \nu b_\tau/2$.

Proposition 3 (Optimal adaptive verification)**.** *Suppose the feasible program set $\mathcal{T}_f(h)$ is nonempty. The optimum*

$$V(h;\lambda) = \max_{\mathbf{w}\in\Delta(\mathcal{T}_f(h))} \left\{ \sum_\tau w_\tau \widetilde{G}_\tau - C_a\left(\sum_\tau w_\tau a_\tau\right)\right\} \tag{25}$$

exists, and one optimum uses at most two pure programs. For two programs with different coverage rates and slope

$$s_{\tau\sigma} = \frac{\widetilde{G}_\sigma - \widetilde{G}_\tau}{a_\sigma - a_\tau},$$

the best coverage on their segment is

$$a^*_{\tau\sigma} = \operatorname{clip}\left(\frac{s_{\tau\sigma} - c_a}{d_a}, \min\{a_\tau, a_\sigma\}, \max\{a_\tau, a_\sigma\}\right). \tag{26}$$

Comparing all pure endpoints and segment optima solves the complete program problem.

At fixed average coverage, the organizer compares how much each possible record would improve settlement. Two probability constraints determine the program mixture, which explains the two-program support. The conditional programs make an economically different choice from either independent or paired checking: they use the first resource report to decide whether further information is worth buying.

After a first report, the second check is valuable when it permits a sufficiently useful change in the resource arrangement. Its value can therefore depend on the first person's reported exposure, even though the two exposures are statistically independent. This dependence arises through the settlement decision. Kattwinkel and Knoepfle (2023) also study the interaction between access to information and costly verification; here the shared interface is what connects the separate checks.

The numerical example uses the primitives in Table 2. Unless stated otherwise, coupling is finite, $\lambda = 2$, and productive investment is inactive, $\varphi = 0$. Welfare in the numerical tables excludes the common trial-output constant $Q_A + Q_B = 24$.

Table 2: Baseline primitives and computational constants.

Parameter	Value
p_A, p_B	2/5 / 2/5
q_A, q_B	1/2 / 1/2
v_H, v_L	3 / 1
η, ℓ	1 / 18
r_-, r_+, b	1 / 3 / 2
ζ, χ	1/10 / 1/10
k_{0A}, k_{0B}	9/5 / 9/5
k_{1A}, k_{1B}	4/5 / 4/5
κ	4
λ	2
c_a, d_a	1/5 / 4/5
ν	0
ε	4/5
ρ	1; noisy comparison: 19/20
Q_A, Q_B	12 / 12
Positive-signal floor $\underline{k}$	1/100000000
Class primal-dual tolerance	1/10000000000

Note: The main policy calculation uses finite coupling and zero productive-investment coefficient.

The baseline illustrates why verification and settlement work together. At $x_A = x_B = 1/2$, the residual resource coefficient is 11/20. An unchecked low type with high resources then has a willingness to pay for continuation that exceeds that of the high type with low resources by 1/5 in continuation value. The ordering needed for exact selection fails. Once both quantities are verified, that sorting conflict disappears. The largest required signal is 53/25, below the available capacity 11/5. Verification at zero settlement would still require a signal of 16/5, above capacity 9/5. Checking resources makes the intermediate settlement implementable; settlement makes the signal physically feasible.

Table 3 shows the procedure responding to that adjustment link. At moderate coupling, checking A first and checking B after a low report gives the organizer room to adapt without paying for both records in every checked case. Near common settlement, a single unchecked module can keep the joint adjustment constrained, which makes paired checking more attractive. Symmetry permits corresponding programs with B checked first; the table displays one optimal support.

Table 3: Coupling and the optimal verification program.

λ	$J(\lambda)$	Net welfare	Coverage	One support
0	0.00000000	0.22947976	45.3650%	$\{00, 10\}$
1/10	0.00691130	0.22837864	45.0606%	$\{00, A_\}$
1	0.04940209	0.22163309	43.1488%	$\{00, A_\}$
2	0.07580811	0.21738006	41.8986%	$\{00, A_\}$
10	0.13948459	0.20642831	38.4928%	$\{00, A_\}$
100	0.16937946	0.20145133	36.8412%	$\{00, 11\}$
∞	0.17336544	0.20095004	36.6707%	$\{00, 11\}$

Note: The infinity row imposes common settlement. Program codes are defined in equation (10); symmetry may yield equivalent optimal supports.

4.2 Verification correlation and realized workload

Paired checks can have an information advantage without a cost advantage. At equal individual coverage a, paired checking and independent checking differ by

$$V_{\text{pair}} - V_{\text{ind}} = a(1-a)\left[J(\lambda) - \frac{\nu}{4}\right]. \tag{27}$$

The information difference is $a(1-a)J$. The paired policy concentrates workload, which adds $\nu a(1-a)/4$ to expected cost. Both policies buy the same average service capacity. Equation (27) compares these two policies at the same marginals; the optimal conditional program can dominate both.

A first check can therefore justify stopping for two separate reasons: the remaining information has little settlement value, or a second check imposes too much additional workload. The full program comparison in Proposition 3 accounts for both. A positive paired-versus-independent difference alone leaves room for a conditional program to perform better.

4.3 Endogenous modularization

Verification can change whether an organizer wants a separable project. Suppose a redesign before the trial reduces coupling from $\bar{\lambda} > 0$ to $\bar{\lambda} - u$, where $u \in [0, \bar{\lambda}]$. The real redesign cost is mu, divided equally between participants. This choice keeps the continuation game fixed and changes the cost of adapting its resource modules.

Theorem 3 (Verification and modularization)**.** *Assume participation holds for every* $\lambda \in [0, \bar{\lambda}]$*. With linear redesign cost, the problem*

$$\max_{0 \le u \le \bar{\lambda}} \{V_0(\bar{\lambda} - u) - mu\} \tag{28}$$

has an endpoint optimum. Complete modularization is strictly preferred to no redesign if and only if

$$m < \frac{V_0(0) - V_0(\bar{\lambda})}{\bar{\lambda}}. \tag{29}$$

The reverse strict inequality selects no redesign; equality makes both endpoints optimal.

For every fixed implementable policy, welfare is affine in the coupling parameter. Optimizing policies makes the value function convex. Subtracting a linear redesign cost preserves the endpoint comparison. Table 4 gives a case where verification creates positive value for modularization, while the no-verification settlement is already identical across participants.

Table 4: Verification can induce complete modularization.

Object	Value
Initial coupling	2
Unit redesign cost	1/250
Redesign-cost threshold	0.006049849242
No-redesign net value	0.21738006
Full redesign net of its cost	0.22147976
Gross redesign gain with verification	0.01209970
Gross redesign gain without verification	0.00000000

Without verification, the symmetric project chooses the same settlement for both participants, so removing the interface cost changes nothing. Resource records allow the organizer to tailor the two settlements. Modularity then makes this tailoring cheaper and becomes worth purchasing. This is the option value of separate adaptation emphasized by Baldwin and Clark (2000), generated here by the information the organizer obtains.

5 Information timing and tolerance for team errors

5.1 Exact selection and early records

Earlier information changes investment incentives before it changes final consent. A favorable record about one participant raises the other participant's probability of meeting a high type. In the interior of the capacity-bound formula, the required settlement responds according to

$$\frac{\partial e_i^k}{\partial p_j} = \frac{(a_L + b\overline{r}_i)k_{1i} + \gamma b\overline{r}_i k_{0i}}{(k_{1i} + p_j \gamma b\overline{r}_i)^2} > 0, \tag{30}$$

when the numerator is positive. A favorable report about B makes B a more attractive partner for A's low type. Preventing imitation then requires a larger signal from A. Once that signal reaches

the activity-window limit, the organizer must release more of A's resource commitments. Earlier information can therefore raise the cost of exact selection even though it is useful for evaluating the partner.

For each release schedule, the organizer optimizes verification after every early history. Settlement is then optimized after the resource reports. The expected value of a schedule is $\sum_h \Pr(h \mid R)V(h;\lambda)$. Table 5 compares these fully reoptimized policies. Delayed disclosure is best for the exact target at the baseline capacity.

Table 5: Optimized disclosure timing under exact reciprocal continuation.

Schedule	Net welfare	Welfare including trial
00	0.21738006	24.21738006
10	0.13985763	24.13985763
01	0.13985763	24.13985763
11	0.07274611	24.07274611

Note: 0 denotes late release and 1 early release; the first position refers to A. Every public history is reoptimized.

Every schedule in Table 5 implements continuation with probability $p_A p_B = 4/25$. Table 6 explains why their costs differ. The organizer responds to early favorable records by settling more resources, which reduces the signal expenditure needed for selection. That saving is outweighed by higher settlement expenditure and the loss of resources usable through the continuing team. The timing comparison reflects this substitution between settlement and investment.

Table 6: Exact-target welfare by payoff component.

Payoff component	Both late	Both early
Base team benefit	1.28000000	1.28000000
Relationship-bound resources	0.71736512	0.42740333
Independently usable resources	0.39071867	0.51506484
Less investment cost	1.54328974	1.15510216
Less settlement and interface cost	0.47339731	0.82570185
Less verification service cost	0.15401668	0.16891805
Less realized workload cost	0.00000000	0.00000000
Net welfare	0.21738006	0.07274611

Note: Costs are displayed as positive amounts to be subtracted. Trial output is common and omitted.

The timing comparison depends on the activity window. With unlimited investment capacity, an early-release policy can copy any delayed verification and settlement policy. The sorting constraint is independent of the posterior probability of a high counterpart. Conditional welfare at a fixed resource policy is affine in $p_A(h)p_B(h)$, whose expectation is $p_A p_B$. The copied early policy therefore attains the delayed value, and optimizing it can only improve that value. Appendix A.7 gives the argument. This is a model-specific use of the distinction between information comparisons and the decision problem in which information is used (Blackwell 1953).

A wider finite activity window can reverse the balance. In the example in Appendix B, early records permit valuable policy adjustments without forcing as much additional settlement. Early release

then performs better after reoptimization. The comparison locates the information-timing effect in the strength of the incentive constraint, which complements the relational-incentive perspective in Kolotilin and Li (2021).

5.2 Why a noisy check can have zero exact-target value

A highly accurate report can leave the binding private states unchanged. Under the baseline primitives and delayed continuation records, every report with $\rho < 1$ retains both resource types. The exact sorting requirement is $x_i \geq 5/9$. At $(5/9, 5/9)$, both separate objective derivatives are negative and the interface gradient is zero. Hence this pair is the exact-target optimum at every resource-report history, for every finite $\lambda \geq 0$.

At this fixed settlement, conditional welfare is affine in the posterior resource means. Averaging over the complete set of records restores their prior means, so the information cannot improve expected settlement value. Any positive verification cost then makes zero coverage optimal. An error-free check changes the possible resource types and can relax the constraint itself. Appendix A.7 formalizes this support distinction.

The exact target thus makes the rarest possible imitator decisive. A small error allowance changes that trade-off: the organizer can accept some misclassification in exchange for lower signaling or settlement costs. Probabilistic verification and the distinction between acquisition and screening help frame this choice (Ball and Kattwinkel 2025; Celik and Strausz 2026). We evaluate each permitted error at the true payoffs of the team it creates or dissolves.

5.3 Cutoff equilibria with final refusal

Let the permitted total error be δ. The unrestricted design value is

$$V_\delta = \sup_{\mathcal{D}, e \in PBE(\mathcal{D})} \{W(e) : \Pr_e(D \neq \mathbf{1}_{HH}) \leq \delta\}. \tag{31}$$

We obtain a feasible value and a certified optimum within an explicit class $\mathcal{C}$. At a given public history, choose a nonempty proper subset S_i of each participant's positive-probability private types. Members invest a common positive amount k_i; other types invest zero. The class requires all positive-group members to consent to a positive-group counterpart after every possible late record. Every own type must prefer to refuse a zero-group counterpart after every such record. Indifference is resolved toward consent within the positive group and toward refusal of the zero group.

A minimum positive cost $\underline{k} = 10^{-8}$ closes the computational strategy class. This floor restricts the candidate class only; deviations still range over the original continuous action set $[0, K_i(x_i)]$. Define

$$\alpha_i = \Pr(S_i), \quad h_i = \Pr(H_i \cap S_i), \quad \mu_i = \frac{h_i}{\alpha_i}, \quad \mu_i^0 = \frac{p_i - h_i}{1 - \alpha_i}, \qquad 0 < \alpha_i < 1. \tag{32}$$

Final refusal affects the incentive to copy a positive signal. For a true type $t_i = (\theta_i, r_i)$, let $\mu_j(s)$ be the posterior high-type probability in the counterpart's positive group after late record s. The opportunity value of copying the positive signal is

$$\mathcal{B}_i(t_i; x_i) = \alpha_j \sum_s \Pr(s \mid S_j) \left[v_{\theta_i} + (1 - \gamma x_i) b r_i + g(\mu_j(s)) \right]_+ , \tag{33}$$

where $g(\mu) = \eta\mu - \ell(1-\mu)$. If the counterpart's record is already public, the sum has one term. The positive part represents a choice: after a bad record, the participant can refuse; after a good record, it can accept. Computing that choice before averaging gives the correct incentive to invest.

Proposition 4 (Cutoff implementation)**.** *Maintain the positive-group and zero-group consent conditions defining $\mathcal{C}$. A pair of positive signals is supported by a PBE in this class whenever*

$$\max \left\{ \underline{k}, \max_{t_i \notin S_i} \mathcal{B}_i(t_i; x_i) \right\} \leq k_i \leq \min \left\{ K_i(x_i), \min_{t_i \in S_i} \mathcal{B}_i(t_i; x_i) \right\}. \tag{34}$$

Continuation then occurs precisely when both participants belong to their positive groups. Its false-positive and false-negative probabilities are

$$FP = \alpha_A \alpha_B - h_A h_B, \qquad FN = p_A p_B - h_A h_B. \tag{35}$$

Every member of the positive group consents at each late history, so A's proposal adds no selection within that group. Excluded types are evaluated differently: a deviator may pay the positive signal and then refuse following a bad record. Equation (33) includes that option. Appendix A.8 specifies the supporting beliefs and checks the remaining actions.

5.4 Continuous settlement and the welfare frontier

The finite type space makes the cutoff class tractable. Each participant has at most four types, which gives at most fourteen nonempty proper subsets. For each subset pair, the algorithm partitions settlement at crossings of private continuation values and at the activation points of the positive-part terms in equation (33). The lower signal bound, including $\underline{k}$, is affine within each interval. The consent and incentive inequalities then define intervals for the two settlement choices.

The resulting welfare problem remains a concave quadratic on a rectangle. If the minimum signal in a region is $k_i = a_i^k + b_i^k x_i$, and $r_i^S = E[r_i \mathbf{1}_{S_i}]$, it has the form

$$F_z^S(\mathbf{x}) = c_z^S + \beta_{Az}^S x_A + \beta_{Bz}^S x_B - C_\lambda(\mathbf{x}), \tag{36}$$

$$\begin{aligned} c_z^S &= \sum_i \alpha_j E[(v_{\theta_i} + g(\theta_i)) \mathbf{1}_{S_i}] + b \sum_i \alpha_j r_i^S - \sum_i \alpha_i a_i^k, \\ \beta_{iz}^S &= \zeta b m_i - \gamma b \alpha_j r_i^S - \alpha_i b_i^k. \end{aligned} \tag{37}$$

The allocation determines the true continuation benefits of each group, and the signal cost determines what it takes to sustain that allocation. Each subset pair contributes its highest welfare across the continuous settlement choices. The zero-investment no-continuation equilibrium is also included, so the organizer can compare selective continuation with ending the partnership.

Theorem 4 (Error-tolerance certificates)**.** *Allow public randomization over the four release schedules and over the defined verification programs. For an error price $s \geq 0$, let*

$$H(s) = \sup_{\mathcal{D}, e \in \mathcal{C}(\mathcal{D})} \{W(e) - s \operatorname{Err}(e)\}. \tag{38}$$

Optimizing the finite cutoff candidates at each resource-report leaf, followed by program optimization and the choice of release schedule, yields $H(s)$. Consequently,

$$V_\delta^{\mathcal{C}} \leq H(s) + s\delta. \tag{39}$$

A public mixture of feasible policies supplies a lower bound. Together, the two bounds certify the optimum within $\mathcal{C}$.

The error price is the marginal value assigned to tighter selection. For a given price, the organizer compares settlement and verification programs using welfare net of the expected error charge. Varying the price traces the best available trade-off. The computation brackets the price with exact rational arithmetic and retains feasible policies on both sides of a binding error budget. The primal-dual certificate follows the optimization logic in Boyd and Vandenberghe (2004); the equilibria themselves come from Proposition 4.

A full-information benchmark measures the remaining distance to costless selection. Under the stronger condition

$$\ell > v_H + v_L + \eta + 2br_+, \tag{RS}$$

a full-information relaxation bounds every PBE of the main $\varphi = 0$ model. It makes types and resources public for free. It removes signaling and verification costs and drops incentive constraints. Non-HH continuation then has negative total incremental value. In HH states, $\zeta \leq \gamma$ makes zero settlement optimal. Let

$$G_{11}^{NC}(\lambda) = E_{r_A, r_B} \max_{x \in [0,1]^2} \{\zeta b(r_A x_A + r_B x_B) - C_\lambda(\mathbf{x})\}.$$

The unrestricted upper bound is

$$U_{FI} = 2p_A p_B a_H + p_A p_B bE[r_A + r_B] + (1 - p_A p_B) G_{11}^{NC}(\lambda). \tag{40}$$

The cutoff optimum satisfies $\underline{V}_\delta^{\mathcal{C}} \leq V_\delta^{\mathcal{C}} \leq \overline{V}_\delta^{\mathcal{C}}$. The unrestricted value instead lies in $\underline{V}_\delta^{\mathcal{C}} \leq V_\delta \leq U_{FI}$. The first interval certifies the optimum in the defined class; the second uses the full-information relaxation for all PBE.

Tables 7 and 8 trace the cutoff frontier as the error budget rises. The full-information upper bound is 2.6384 throughout. At a 0.5-percent budget and 95-percent report accuracy, the feasible welfare value is approximately 0.26559; the class gap is below 10^{-10}.

The first errors admitted in these examples are false-positive continuations. Allowing them reduces the investment needed to deter every low-value type, while all high-value pairs can still continue. At larger budgets, the selected groups also exclude some high-value pairs. Early records then help the organizer choose among these less costly allocations. The optimal policy moves from delay toward early release, with public mixtures at intermediate budgets.

Table 7: Error-tolerance frontier with 95-percent resource accuracy.

δ	Feasible value	FP	FN	$\Pr(D=1)$	Class gap	Schedule
0.00%	0.14716049	0.0000%	0.0000%	16.0000%	0.00e+00	00
0.10%	0.17300209	0.1000%	0.0000%	16.1000%	1.52e-11	00
0.50%	0.26559158	0.5000%	0.0000%	16.5000%	2.95e-11	Mix
1.00%	0.38132813	1.0000%	0.0000%	17.0000%	2.34e-11	Mix
2.50%	0.72853781	2.5000%	0.0000%	18.5000%	5.08e-12	Mix
5.00%	1.28627705	5.0000%	0.0000%	21.0000%	1.34e-11	11
10.00%	1.50471975	6.0480%	0.6080%	21.4400%	0.00e+00	11
16.00%	1.50471975	6.0480%	0.6080%	21.4400%	0.00e+00	11

Note: Class gap is the upper bound minus feasible welfare within the cutoff PBE class. The all-PBE bound is 2.6384. Mix denotes a public lottery over schedules. Trial output 24 is omitted.

Table 8: Error-tolerance frontier with error-free resource verification.

δ	Feasible value	FP	FN	$\Pr(D=1)$	Class gap	Schedule
0.00%	0.21738006	0.0000%	0.0000%	16.0000%	0.00e+00	00
0.10%	0.24100612	0.1000%	0.0000%	16.1000%	3.22e-12	Mix
0.50%	0.33551034	0.5000%	0.0000%	16.5000%	1.61e-11	Mix
1.00%	0.45355455	1.0000%	0.0000%	17.0000%	6.46e-13	11
2.50%	0.80299619	2.5000%	0.0000%	18.5000%	4.94e-11	11
5.00%	1.35109189	5.0000%	0.0000%	21.0000%	5.03e-11	11
10.00%	1.45494854	5.1600%	0.4800%	20.6800%	0.00e+00	11
16.00%	1.45494854	5.1600%	0.4800%	20.6800%	0.00e+00	11

Note: Class gap is the upper bound minus feasible welfare within the cutoff PBE class. The all-PBE bound is 2.6384. Mix denotes a public lottery over schedules. Trial output 24 is omitted.

The accuracy comparison also depends on the feasible posterior groups. Improving a report can change which groups satisfy uniform consent after every late record. The two tables therefore describe separate optima of the cutoff class; the broader information-design comparison is discussed in Section 7.1.

6 Productive investment and implementation limits

6.1 Joint output feeds back into imitation

Productive activity can alter the cost of selecting a team. Team-incentive models make production and incentive compatibility jointly relevant (Holmström 1982). Joint-project models also show why the timing and commitment of contributions matter (Admati and Perry 1991). In this extension, the technological term $\varphi y_A y_B$ is realized before the continuation decision. A participant who imitates a high type earns the same production return from the counterpart's investment.

Take $\varphi > 0$ and impose $\varphi \max_i K_i(x_i) < 1$ in the analyzed settlement region. Marginal private investment remains costly even when the counterpart invests at capacity. Under the exact HH target, low types still invest zero, and all resource subtypes of the high continuation type use a common signal. Define $c_i = \varphi p_j/2$.

Proposition 5 (Productive investment)**.** *At a fixed public history and settlement vector satisfying the low-productivity condition, exact HH implementation requires*

$$L_i \le k_i(1 - c_i k_j) \le H_i, \qquad 0 < k_i \le K_i. \tag{41}$$

When $L_i \le H_i$ for both participants, these inequalities have a solution if and only if the least nonnegative fixed point

$$k_A = \frac{L_A}{1 - c_A k_B}, \qquad k_B = \frac{L_B}{1 - c_B k_A} \tag{42}$$

lies within the capacity rectangle. In the symmetric case, under $\varphi pK < 1$, the condition reduces to

$$L \le K(1 - \varphi pK/2). \tag{43}$$

Production reduces the private net cost of every positive investment, including an investment made to imitate. Deterring a low type can therefore require a larger physical input. At the least fixed point, the extra expected output offsets the corresponding increase in signaling expenditure, so welfare at a given settlement retains equation (19). The substantive change is the settlement needed to fit the new signal within capacity.

Table 9 shows that change in the baseline high-resource cell. At $x_A = x_B = 1/2$, the critical production coefficient is $10/121$. A coefficient of $1/10$ lies above it: even the largest feasible signal is too cheap in net terms to prevent imitation. Increasing settlement lowers the resource return to imitation and releases capacity, which restores exact continuation.

Joint production ties each person's signaling cost to the counterpart's input. It therefore changes the incentive problem even when the continuation benefits themselves remain fixed. Neckebrouck and Zellweger (2026) examine how team objectives interact in a computational model of entrepreneurial decisions. Here the interaction works through a precise margin: productive investment affects the physical signal required to screen a partner.

Table 9: Productive investment changes the implementation boundary.

φ	x_i	k_i	K_i	Test slack	Valid	Welfare
0	1/2	53/25	2.20000000	0.08000000	Yes	0.74000000
1/10	1/2	11/5	2.20000000	-0.01680000	No	n/a
1/10	457/900	11/5	2.20622222	0.00567389	Yes	0.73365679

Note: Both resource exposures are verified as high. The capacity test uses the best feasible signal under the stated productivity. The failed candidate has no equilibrium welfare entry.

6.2 Participation and strong implementation

Trial participation must be consistent with the continuation policy. A participant can always invest zero and refuse to continue. For $\varphi = 0$, a uniform lower bound on trial utility is

$$Q_i - \frac{\kappa}{4} - \frac{\lambda}{4} - \frac{C_a(1)}{2} - \frac{\nu}{4}. \tag{44}$$

If this exceeds M_i, every type strictly enters regardless of beliefs about the counterpart or the continuation equilibrium. B accepts any invitation, and A anticipates this response. Participation therefore leaves the prior unchanged. The baseline bounds are in Table 10.

Table 10: Uniform participation guarantees at the baseline coupling.

Participant	Trial guarantee	Outside bound	Strict margin
A	51/5	10	1/5
B	51/5	10	1/5

The same guarantee applies to the low-productivity extension because zero investment generates no joint output. Under its productivity condition, direct-channel investment cannot raise utility above the stated outside bound. A modularization fee must be deducted from the individual's guarantee. Large finite-coupling entries in Table 3 describe the post-entry design problem where this conservative entry bound fails; the baseline $\lambda = 2$ passes it. At the common-settlement limit, the interface mismatch is zero and a tighter entry bound applies.

Entry and continuation require different forms of coordination. Trial output can make participation attractive to every type, as commitment can support relationship development in Watson (2002). The organizer must still rely on voluntary consent afterward. The next proposition examines the stronger requirement that every equilibrium deliver the target allocation, the distinction emphasized in implementation theory (Maskin 1999).

Proposition 6 (Failure of small-error strong implementation)**.** *Maintain condition (R), full-support continuation records and the stated final-consent protocol. In the full-entry region, every executable verification and settlement policy admits a no-contact PBE. Its error probability is $p_A p_B$. Therefore no such policy strongly implements an error target $\delta < p_A p_B$. Moreover,*

$$\sup_{\mathcal{D}} \inf_{e \in PBE(\mathcal{D})} W(e) \leq \sup_{\mathcal{D}} W_{NC}(\mathcal{D}). \tag{45}$$

The no-contact construction remains a PBE with the specified joint production technology.

The obstacle is coordinated refusal. A unilateral investment cannot create joint output when the other participant invests zero, and a proposal cannot secure cooperation when it is rejected. Resource verification leaves these responses feasible. In the baseline, the best no-contact policy yields net welfare 0.08, so equation (45) places the worst-equilibrium design value at or below that benchmark.

7 Discussion and conclusion

7.1 Institutional scope and equilibrium selection

The model concerns a fixed pair whose resource arrangements can be changed before voluntary continuation. The organizer commits to resource settlement and record release, with bounded restoration afterward. The entry condition keeps the pair's type distribution fixed. Search for other partners and selection into the trial are separate organizational choices. Likewise, allowing arbitrary consent-contingent transfers or unrestricted renegotiation would change the available screening instruments.

The equilibrium claims have a precise boundary. Exact reciprocal continuation has a complete implementation characterization. Error-tolerant optimization covers the defined cutoff class, whose uniform-consent restriction makes every feasible allocation explicit. The full-information benchmark bounds all PBE, and Proposition 6 establishes the coordination problem that prevents a small-error guarantee across equilibria. Equilibrium refinements such as those in Cho and Kreps (1987) and Banks and Sobel (1987) require their own analysis in this bilateral game; sequential equilibrium additionally imposes consistency of beliefs (Kreps and Wilson 1982).

The information choice consists of four release schedules and automatic resource checks. The cutoff class does not include every garbling of those checks, so its values across accuracies are not an unrestricted Blackwell comparison. Productive investment is characterized in the stated low-productivity region; the main policy frontier uses purely costly investment. The numerical examples apply one common utility scale to the declared model environments.

7.2 Conclusion

Resource architecture changes the value of learning about a prospective partner. A check affects the settlement that makes further investment informative. When the modules are separable, these adjustments contribute additively. An interface cost links them: close to the separable benchmark, the interaction between checks is the product of their average settlement changes. The sign of that product determines local complementarity or substitution.

The same adjustment channel guides the organization of verification. A first report can make a second check worth purchasing, while strong coupling can favor checking both participants together. Verification may also make a modular redesign worthwhile by creating opportunities for separate adaptation. These choices depend on what happens after information arrives, as well as on the precision or price of the information itself.

Error tolerance extends this logic to the continuation decision. Exact selection can require substantial settlement when early good news raises imitation incentives. Allowing some errors permits the organizer to use early records for less costly allocations. Joint production adds a related feedback because imitators receive its return too. Project design and information policy thus jointly determine the cost of sustaining voluntary cooperation.

A Proofs

A.1 Exact implementation

Proof of Proposition 1. Fix a positive-probability public history and a settlement vector. Trial output, settlement expenditure and independent resource value are constants for the subsequent investment problem. A low continuation type never forms a relationship under the target. Its continuation payoff is therefore minus its expected investment cost. It can choose zero and refuse final consent, so optimality requires zero cost almost surely.

Every high type must continue with every high counterpart and refuse every low counterpart. On almost every signal in a high type's equilibrium support, its expected relationship value is consequently $p_j[a_H + t_i b r_i]$. A mixed strategy puts positive weight only on best responses. All its supported costs must therefore be equal. This excludes a nondegenerate cost distribution for a fixed private type.

Resource subtypes with high continuation value can copy one another's complete investment and consent plans. The counterpart's benefit depends on continuation type, and the exact target includes all resource combinations. Copying another high subtype's plan thus gives the same probability p_j of obtaining a high counterpart. A subtype paying a larger signal cost would strictly prefer the cheaper plan. All high subtypes use a common k_i.

The low type with exposure $\overline{r}_i$ can copy this plan. Against a high counterpart, its true relationship increment is $a_L + t_i b \overline{r}_i > 0$, so it will consent. Against a low counterpart it can refuse. Its deviation payoff is $L_i - k_i$, which implies $k_i \geq L_i$. The high type with exposure $\underline{r}_i$ can instead invest zero and refuse. Hence $k_i \leq H_i$. Physical feasibility gives $k_i \leq K_i$.

For sufficiency, choose each k_i from the stated interval. High types invest k_i and low types invest zero. The designated positive signal implies a high type by Bayes' rule; zero implies a low type. Assign a low-type belief after any other investment. A proposes only when its own designated signal will be accepted and B has the designated signal. B accepts an offer following A's designated signal. Condition (R) makes these consent decisions optimal for every own type, including a low type who has copied the positive signal.

Each high type obtains $p_j[a_H + t_i b r_i] - k_i \geq 0$. Any other investment yields no relationship and a nonpositive payoff. Each low type's best imitation payoff is no greater than $L_i - k_i \leq 0$. Other positive investments only incur cost. These arguments cover the full continuous investment interval.

Late records have strictly positive likelihood under each continuation type, so they preserve the path's pure type inference and cannot invalidate the imitation argument. Specifying the same beliefs at unused consent histories supports the stated refusal rules. Thus the strategies and beliefs form a PBE. □

A.2 Settlement and its welfare account

Proof of Proposition 2. Divide the sorting condition $L_i \leq H_i$ by $p_j > 0$. It becomes $[1 - \gamma x_i]b(\overline{r}_i - \underline{r}_i) \leq \Delta$. A singleton support satisfies it automatically. Otherwise, for $\gamma > 0$, its lower bound is e_i^s. The capacity condition rearranges to

$$p_j(a_L + b\overline{r}_i) - k_{0i} \leq (k_{1i} + p_j\gamma b\overline{r}_i)x_i.$$

This gives e_i^k when the coefficient is positive. A zero coefficient is checked without division. Intersecting the inequalities with $[0, 1]$ gives the stated rectangle or an empty feasible set.

The target allocation fixes all relationship benefits at a given settlement. Raising either signal only increases real cost, so $k_i = L_i$ maximizes target welfare. The HH probability is P_z. The expected base relationship gain is $2P_z a_H$. Bound resources yield $P_z b \sum_i t_i m_i$, and independent resources yield $\zeta b \sum_i x_i m_i$. Expected signal costs are $P_z \sum_i [a_L + t_i b\overline{r}_i]$. Adding these terms and subtracting C_λ yields equation (19). Resource gains are not dropped from welfare; their cancellation with a component of signaling expenditure is algebraic.

The eigenvalues of $\mathbf{Q}_\lambda$ are $\kappa/2$ and $\kappa/2 + 2\lambda$. Both are positive. Strict concavity and compact feasibility imply existence and uniqueness. An interior optimum solves $\mathbf{Q}_\lambda \mathbf{x} = \boldsymbol{\beta}$. On each boundary, the restricted objective is a concave univariate quadratic. Comparing its stationary point, when feasible, with boundary endpoints and the interior candidate finds the unique global optimum. □

For a computational optimality certificate, write the box as $\boldsymbol{l} \leq \mathbf{x} \leq \boldsymbol{u}$. At a candidate optimum, the gradient must vanish at an interior coordinate, be nonpositive at a lower-bound coordinate and be nonnegative at an upper-bound coordinate. Concavity makes these sign conditions sufficient. The independent checker enumerates the nine combinations of lower, free and upper status for the two coordinates.

A.3 The verification envelope

Proof of Proposition 3. A pure program either stops immediately or begins by checking A or B. Following a first check, its two possible reports each lead to stopping or checking the other participant. The four deterministic continuation rules are: stop after either report, continue after either report, continue only after a low report, or continue only after a high report. Merging the two orders of unconditional double verification gives eight distinct final-information programs. Randomized continuation can be selected by an initial lottery over these pure rules.

Fix a feasible expected coverage a. Program weights satisfy $\sum_\tau w_\tau = 1$ and $\sum_\tau w_\tau a_\tau = a$. The value before the service fee is linear in these weights. An extreme point has at most two positive components, since three vectors $(1, a_\tau)$ would be linearly dependent and permit a nonzero feasible perturbation. Thus an optimum at fixed coverage lies on a segment between two program points, or at a pure point. Equal-coverage points can be replaced by the one with the largest value.

On a segment, the net objective is affine gross value minus $c_a a + d_a a^2/2$. Its curvature is $-d_a < 0$ and its projected stationary point is equation (26). Finite comparison of every feasible segment and endpoint therefore gives the global optimum. The nonempty finite program set and compact simplex ensure attainment. □

The tangent certificate offers a separate check. At chosen coverage a^*, let $\omega = c_a + d_a a^*$. Every support program must maximize $\widetilde{G}_\tau - \omega a_\tau$. By convexity, $C_a(a) \geq C_a(a^*) + \omega(a - a^*)$. Hence every alternative mixture has value at most

$$\max_\tau (\widetilde{G}_\tau - \omega a_\tau) + \omega a^* - C_a(a^*), \tag{A.1}$$

which equals the chosen value when the support conditions hold. Endpoint coverages satisfy the same inequality. The certificate recalculates the shadow price from actual coverage.

A.4 Local complementarity

Proof of Theorem 1. At $\lambda = 0$, conditional welfare is a sum of two separate settlement problems. The feasible rectangle is also separate. A check of one participant changes only that participant's conditional resource distribution. Independence therefore gives $J(0) = 0$.

The unique optimum in every public cell permits the envelope derivative at zero from the right. The derivative of the optimized cell value is $-(X_A - X_B)^2/2$. Apply this derivative to the four verification programs. In their cross-difference, the terms involving X_A^2 and X_B^2 cancel. The remaining expression is

$$E[X_A^1 X_B^1] - E[X_A^1] X_B^0 - X_A^0 E[X_B^1] + X_A^0 X_B^0.$$

Independence factors the first expectation and gives equation (22). The bounded feasible set and finite resource histories allow differentiation through the finite expectations. A strictly positive or negative right derivative, together with $J(0) = 0$, establishes the corresponding sign on a sufficiently small positive neighborhood. □

The derivative is explicit in the primitives. At the separable benchmark, a cell's optimizer is $\mathrm{clip}(2\beta_i/\kappa, e_i, 1)$. Substitution into equation (22) gives each adjustment term directly. The tabled signs therefore come from the resource distributions and the capacity bounds, without an assumed sign on an information cross-partial.

A.5 The common-settlement limit and workload

Proof of Theorem 2. For every fixed feasible $\mathbf{x}$, its value is affine and nonincreasing in λ. Taking a maximum gives a nonincreasing convex value function. Common settlements are feasible in every finite-coupling problem, so $F_\lambda^* \geq F_\infty^*$.

Let $f_i(x_i) = \beta_i x_i - \kappa x_i^2/4$. On $[0,1]$, $|f_i'(x_i)| \leq B_z$. Given a feasible pair, raise its smaller coordinate to its larger coordinate. This remains in the implementation rectangle and creates a common settlement. The change in the separate objective is bounded by $B_z|x_A - x_B|$. Thus

$$F_\lambda(\mathbf{x}) \leq F_\infty^* + B_z|x_A - x_B| - \frac{\lambda}{2}(x_A - x_B)^2 \leq F_\infty^* + \frac{B_z^2}{2\lambda}.$$

This proves convergence and its rate. Finite averaging across program leaves preserves the limit. The same uniform bound over a finite program set preserves convergence of the optimized mixture. Taking a finite linear combination gives the limit of J. □

For equation (27), paired checking has $\Pr(N = 2) = a$ and $E[(N/2)^2] = a$. Independent checking has $\Pr(N = 2) = a^2$, $\Pr(N = 1) = 2a(1-a)$ and $E[(N/2)^2] = (a+a^2)/2$. The excess paired workload fee is $\nu a(1 - a)/4$. At equal marginals, the gross program-value difference is $a(1 - a)J$, which gives the stated net difference. Workload fees are linear in program weights after each program's leaf probabilities have been evaluated, so Proposition 3 applies to $\widetilde{G}$.

A.6 Module redesign

Proof of Theorem 3. Exact implementation constraints depend on own settlement and resource supports, not directly on λ. Fix a complete resource policy and an exact-target equilibrium. Their welfare is affine in λ. Optimizing over the same feasible policy set produces a convex $V_0(\lambda)$. The required uniform participation condition keeps that feasible set unchanged by entry selection.

Consequently, $V_0(\bar{\lambda} - u) - mu$ is convex in u. Every point of the closed interval is a convex combination of its two endpoints, so its value is bounded above by the same combination of endpoint values. At least one endpoint is optimal. The difference between complete modularization and no redesign is $V_0(0) - m\bar{\lambda} - V_0(\bar{\lambda})$, giving the threshold. A strict difference makes the better endpoint uniquely optimal; at equality both endpoints attain the same maximum. □

A.7 Disclosure and the support of resource reports

Differentiate the untruncated expression for e_i^k in equation (17). The terms containing p_j in the numerator cancel. The result is equation (30). If the lower bound is truncated at zero, its local derivative can instead be zero. A favorable early record can therefore raise the settlement needed to satisfy capacity, but only where that bound matters.

For the unlimited-capacity comparison, take an arbitrary feasible delayed-release policy. Let the early policy ignore its extra record when selecting verification and settlement. Sorting conditions survive because the strictly positive p_j divides out. Investment remains feasible without a capacity limit. The new lowest-cost signals implement the target at each early history. For fixed resource records, conditional welfare is affine in $P_z = p_A(h)p_B(h)$. Independence and iterated expectations give $E_h[P_z] = p_A p_B$. The copied early policy has the delayed policy's ex ante value, including equal verification costs. Optimizing early policies can only raise it.

For the noisy-report exact target, use the baseline primitives and delayed records. Every $\rho < 1$ retains support $\{1, 3\}$, so each sorting lower bound is $5/9$; the capacity lower bound is smaller. The largest individual coefficient β_i over posterior means in $[1, 3]$ is $97/125$, strictly below $(\kappa/2)(5/9) = 10/9$. At $x_A = x_B = 5/9$, the interface gradient is zero and both other gradients point below the feasible rectangle. Strict concavity makes this the unique conditional optimum for any finite λ. Its value is affine in the posterior resource means. All leaves of an adaptive report program form a complete probability partition, so the expected means return to their prior values. All programs therefore have the same gross exact-target value, $298/2025$. Every positive coverage has a positive service cost. Zero coverage is optimal. At $\rho = 1$, verified supports become singletons and this support-preservation argument no longer applies.

A.8 Cutoff strategies and continuous candidate enumeration

Proof of Proposition 4. Fix the public history and the two type subsets. Independence determines the counterpart's conditional type distribution. For each positive or zero investment group, Bayes' rule and the late-record likelihood give its final type posterior.

Consider a type copying the positive signal. The other participant's maintained strategy consents to every possible member and late record of the positive group. The deviator can still refuse after seeing the counterpart's record. Its optimal relationship opportunity is therefore exactly $\mathcal{B}_i$ in equation (33). Every possible own type prefers to reject a zero-group counterpart by the class restriction. Such a path cannot offer an additional profitable relationship.

The upper bound in equation (34) gives nonnegative net utility to every positive-group member. The lower bound makes imitation unprofitable for every excluded type. A different continuous positive signal is assigned a low-type belief and obtains no relationship; its incremental payoff is nonpositive. At positive-probability final histories, all positive-group members consent. A's offer therefore adds no further selection within that group. An unexpected offer after a zero signal can be assigned a low-type belief, and condition (R) makes B's refusal optimal. The specified investments, consent decisions and beliefs form a PBE.

The continuation event is $S_A \cap S_B$, with probability $\alpha_A \alpha_B$. Its intersection with true HH has probability $h_A h_B$. Subtracting this intersection from the continuation probability gives FP, and subtracting it from $p_A p_B$ gives FN. □

For a fixed subset pair, the value $v_\theta + (1 - \gamma x_i) br_i$ is affine in x_i. Its ordering can change at finitely many intersections. On each interval with a fixed ordering, the type outside S_i with the largest opportunity value is fixed, because the positive-part expectation is weakly increasing in that affine value. Late-record terms have finitely many activation points. Including those points and the positive-cost floor partitions the interval into affine signal-cost regions.

The consent inequalities and the individual signal bounds are linear in each region. Their intersection is an interval for each participant, and the joint feasible set is a rectangle. Given the subsets, error probabilities depend on type masses and not on settlement. Adding the two actual payoff functions gives equations (36) and (37). The negative Hessian is again $\mathbf{Q}_\lambda$. Thus the finite enumeration of regions combined with the global rectangle solution exhausts the maintained cutoff class. Equality boundaries retain the stated consent tie rules. This construction does not assume that all possible PBE use these two-signal groups.

A.9 Primal and dual welfare bounds

Proof of Theorem 4. For a fixed error price, a candidate at a resource-report leaf contributes its welfare minus the error price times its error. The organizer observes the leaf before its settlement decision and is committed to a policy mapping. It can select the maximizing candidate independently at every leaf. Expected penalized values then become program values, to which Proposition 3 applies. Early histories are averaged using their actual probabilities and their own purchased service capacities. Taking the maximum across the four release schedules yields $H(s)$; a public mixture cannot exceed this maximum penalized value.

For any policy in the class with error at most δ, $W - s\,\mathrm{Err} \le H(s)$ and $s\,\mathrm{Err} \le s\delta$. Adding

the inequalities proves the upper bound. Each computed feasible policy has an independently verified PBE. Publicly mixing two such policies preserves their PBE strategies conditional on the lottery result. If their errors bracket the budget, weights can attain it. Policies using the same release schedule can additionally merge their conditional coverages; convex service costs make this operation weakly cheaper, without changing the leaf allocations. Different release schedules retain their separate conditional service fees. The resulting policy supplies the claimed lower bound. □

The full-information bound relaxes a different set of constraints. Under (RS), even the largest resource gains leave every non-HH relationship with negative total incremental value. Full information therefore selects no continuation in those states. In HH, the resource gain from independent use has slope ζbr_i, while lost bound resources have slope γbr_i. Since $\zeta \leq \gamma$, settlement weakly lowers total HH welfare, before its nonnegative cost. Thus HH selects continuation and zero settlement. Other states optimize independent resource use over the full box. The independence of resource and continuation types gives equation (40). Free information and removal of incentive constraints enlarge the feasible set; the deletion of signaling and audit costs raises its value. This proves validity for every PBE of the main model.

A.10 Productive investment

Proof of Proposition 5. At the stated low productivity, the own net cost of investment is positive for every feasible counterpart investment. Under exact continuation, a low type earns no relationship return and therefore chooses zero investment. High types receive the same relationship allocation across their resource subtypes and face the same expected marginal investment cost, so they use a common signal.

If the counterpart's high type invests k_j, the expected production return to a unit of own investment is $\varphi p_j k_j/2 = c_i k_j$. A low type copying k_i receives this return too. Its net signal cost is $k_i(1 - c_i k_j)$, which yields the lower inequality in equation (41). A high type's deviation to zero gives the upper inequality. An unused signal still produces output, but its net payoff without a relationship is $-y_i(1 - c_i k_j) < 0$. Thus productive off-path deviations are included.

Suppose a feasible vector $\mathbf{k}$ satisfies these inequalities. The map in equation (42) is increasing on the capacity rectangle and has positive denominators there. It maps $\mathbf{k}$ weakly below itself. Iteration from zero is nondecreasing and bounded by $\mathbf{k}$. It converges to the least fixed point within capacity. At that point both lower incentive bounds bind. The sorting inequalities ensure the upper bounds as well. Conversely, a fixed point within capacity directly satisfies the lower bounds and, under sorting, the upper bounds.

In the symmetric case, $k(1 - \varphi pk/2)$ is increasing on $[0, K]$ under $\varphi pK < 1$. Its range reaches L if and only if equation (43) holds. This proves the stated boundary. □

For a direct calculation of the fixed point, let $z = k_A k_B$. The binding lower bounds imply $k_i = L_i + c_i z$, and

$$c_A c_B z^2 - (1 - c_A L_B - c_B L_A)z + L_A L_B = 0. \tag{A.2}$$

The smaller feasible nonnegative root gives the least fixed point. Its implied signals must still satisfy capacity. When $\varphi = 0$, set $k_i = L_i$ directly. At the least point, expected investment

expenditure minus production equals $p_A L_A + p_B L_B$. Consequently, conditional target welfare retains equation (19); productivity changes the feasible settlement set.

A.11 Entry and the no-contact equilibrium

Zero investment and refusal guarantee zero relationship expenditure and nonnegative independent resource value. Own settlement costs are at most $\kappa/4$. The interface and service costs are bounded by $\lambda/4$ and $C_a(1)/2$, respectively. Individual workload cost is at most $\nu/4$. This gives equation (44). If it exceeds the outside upper bound, B strictly joins following any invitation and A strictly invites. No type is selected out, so the subsequent prior is unchanged. The low-productivity case preserves the guarantee because zero own investment generates zero joint production.

Proof of Proposition 6. Fix an executable public policy. Specify that both participants invest zero and A never proposes continuation. The on-path records have their Bayesian posteriors. At an unexpected proposal, B assigns probability one to A being a low continuation type, while retaining any verified resource value. Condition (R) makes refusal strictly optimal for each own type. Other unused investment histories can be assigned the same low-type continuation belief.

A unilateral change in A's investment or proposal still receives refusal. A unilateral change in B's investment cannot force A to propose. Zero investment therefore remains optimal. With the productive extension, the counterpart's zero investment makes unilateral production zero, so a positive unilateral investment is costly. The strategies form a no-contact PBE. In the full-entry region this also extends to a complete participant PBE; outside that region the statement applies to the continuation subgame.

This equilibrium rejects every HH team, so its error is $p_A p_B$. A smaller error target cannot hold in every PBE. The worst equilibrium of each policy has welfare no higher than this policy's no-contact value. Taking the supremum over policies gives equation (45). □

In the no-contact class, signals are zero and conditional welfare is $\zeta b \sum_i m_i x_i - C_\lambda(\mathbf{x})$. The same global rectangle algorithm solves this objective without exact-selection lower bounds. Under the baseline, its optimized institutional value is 0.08. The complete-information no-contact value in equation (40) is a different object: it uses free verified resource quantities and contributes only in non-HH states.

B Computational definitions and supplementary results

B.1 Counterexamples within the same technology

The local substitution example changes only the stated primitives. It uses $b = 1/5$, $\kappa = 1/2$, $\zeta = 1/2$, $p_A = 1/50$, $p_B = 2/5$, $k_{0A} = 1/5$ and $k_{0B} = 1/100$. Other parameters retain their baseline values. Table 11 reports all displayed coupling values. Its negative interaction can turn nonmonotonically as coupling changes, and optimal coverage is zero throughout these entries.

The finite-capacity disclosure counterexample uses $k_{0i} = 2$ and $k_{1i} = 5$, with common settlement and all other baseline primitives. Each release schedule is fully reoptimized. Table 12 shows that the conditional adaptation value of earlier records can outweigh their additional imitation incentives.

Table 11: A resource-coupling environment with verification substitution.

λ	$J(\lambda)$	Net welfare	Coverage	One support
0	0.00000000	0.19105711	0.0000%	{00}
1/10	-0.00053981	0.19104134	0.0000%	{00}
1	-0.00156773	0.19101296	0.0000%	{00}
2	-0.00155247	0.19100805	0.0000%	{00}
10	-0.00150610	0.19100327	0.0000%	{00}
100	-0.00148818	0.19100206	0.0000%	{00}
∞	-0.00148598	0.19100192	0.0000%	{00}

Note: The infinity row imposes common settlement. Program codes are defined in equation (10); symmetry may yield equivalent optimal supports.

Table 12: A finite-capacity environment favoring earlier records.

Schedule	Optimized net welfare
00	0.23876430
10	0.26033747
01	0.26033747
11	0.29408172

Note: The resource and payoff primitives otherwise equal the baseline. Each policy is reoptimized.

B.2 Reproduction and numerical scope

The computation uses exact rational arithmetic. Analytic active sets solve conditional settlement; tangent inequalities certify the optimal verification mixture. Error-frontier outputs record a feasible policy and its dual bound. Table 13 summarizes the generated objects. The numerical implementation is supplied as Online Resource 1.

Table 13: Generated computational objects.

Object	Records	Scope
Coupling-path records	28	Baseline and substitution case
Settlement cells / private states	252 / 1792	Continuous conditional optima
Cutoff candidates	1506	Subset and interval enumeration
Candidate private-state records	20640	Actual type-dependent payoffs
Candidate incentive records	17760	Includes final refusal
Error-frontier records	32	Class primal and dual values
Separate-closeout records	9	Separable comparison
Productive-investment cases	3	Includes one invalid candidate
Reference-baseline incentive records	1296	Primitive payoff differences

Note: Counts refer to records in the accompanying replication archive.

The replication archive provides the economic primitives in `parameters.json` and the declared computational scenarios in `study.json`. One command regenerates numerical results and the

English manuscript. A separate checker reconstructs payoffs from primitive states without importing the main solver. It also recomputes the deviation opportunity after late records and checks the relation between a saved incentive slack and its underlying payoffs. All failed candidates remain available in the output.

The declared parameter scenarios are exploratory model comparisons. All selected scenarios, including failed candidates, are retained in the replication archive.

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Data and code availability. Online Resource 1 supplies the model-generated results, with the parameter files and source code needed to reproduce them. It includes executable reproduction and verification instructions. The study uses no external observational dataset.

Generative-tool assistance. Generative language-model tools assisted with language editing, grammatical correction, and LaTeX formatting. The author is responsible for the final manuscript.

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